

Artificial intelligence

Laboratory work

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Report

The purpose of this laboratory work is to analyze how the query language affects the number of tokens and the cost of accessing artificial intelligence. To do this, the bank's support service reviewed the same request in three languages, Russian, English and Kazakh. I counted tokens, sent requests to the model, and received the number of tokens used, calculating the cost of requests.

The task itself was designed to use the anthropic api and the claude model, but since I didn't have any anthropic tokens, I adapted the lab work to OpenAI.

The following changes have been made:

The Anthropic API has been replaced by the OpenAI API.

In Part 2, requests to Claude were replaced with requests to the OpenAI API.

Instead of the ANTHROPIC_API_KEY, the OPENAI_API_KEY stored in the local was used in the env file.

The gpt-5.6-luna model was used to execute the queries.

The max_tokens parameter has been replaced with max_completion_tokens, which is required for the OpenAI API used.

Part 2 has been changed to preserve the actual input_tokens and output_tokens values returned by the API.

The o200k_base tokenizer from the tiktoken library was used for preliminary token counting.

Part 3 was rewritten because the original version calculated the cost of Claude models. In the adapted version, the cost of the used OpenAI model was calculated.

The measurements file. The json was saved with the results of real queries.

The structure has been preserved, I just changed some points to fit my key type.

In the first part, I counted tokens without sending model requests, I examined three types of text, a short sentence, a customer complaint, and a system prompt

Results:

```
counting tokens with o200k_base (free, no model run)
sentence      en=12 ru=18 kk=21
complaint     en=59 ru=80 kk=118
system_prompt en=40 ru=49 kk=67
request       en=99 ru=129 kk=185 (system + complaint)

answering the same complaint through OpenAI using gpt-5.6-luna:
```

Here we can see that the same meaning occupies a different number of tokens depending on the language.

It turns out that the Russian query contains about 1.3 times more tokens than the English one, and the Kazakh one contains about 1.87 times more

In the second part, after preliminary calculation, I sent the same request to the model in three languages, I used the gpt-5.6-lune OpenAI api, the content was the same, but the feature of the task was that the client mentions the contract and the note, but the model documents themselves were not provided, and as a result, the model indicated that without a contract and bank statements it is impossible to verify the reason for the rate change

```
"one_request_billed": {  
  "en": {  
    "input_tokens": 109,  
    "output_tokens": 119  
  },  
  "ru": {  
    "input_tokens": 139,  
    "output_tokens": 126  
  },  
  "kk": {  
    "input_tokens": 195,  
    "output_tokens": 223  
  }  
}
```

The difference from the preliminary calculation is that these values were obtained from the API after executing real requests.

It turns out that compared to English, Russian used about 1.28 times more tokens, and Kazakh used about 1.79 times more.

Based on this, we can conclude that the Kazakh request not only had more input tokens, but the model's response was much longer.

In the third part, I calculated the cost of one request.

In the input data, \$0.20 per 1 million tokens, and in the output, \$1.20 per 1 million tokens

	EN	RU	KK
input tokens	109	139	195
output tokens	119	126	223
cost, USD	0.000165	0.000179	0.000307
cost, cents	0.0165	0.0179	0.0307

It turns out that one request in English costs about 0.000165 dollars, in Russian 0.000179, and in Kazakh 0.000307

To calculate it for a year, I took into account 2000 requests per day.

The formula came out as the cost of one request multiplied by 2000 and multiplied by 365
Here's results:

	EN	RU	KK
yearly cost	120.16	130.67	223.82

The Kazakh language turned out to be the most expensive with the same number of requests due to the large number of input tokens and the long response.

In this screenshot, you can see two different indicators, namely the ratio of the number of input tokens and the ratio of the total cost of the request.

As a conclusion, I can say with confidence that the same text may require a different number of tokens depending on the language, that is, the cost of a request does not depend only on the number of characters or words in the text, but also on the language, tokenization and length of the model's response, and I also realized that comparing only the number of input tokens is not enough for cost estimates, it is also important to consider the output tokens.